GOVERNMENT ENTERPRISE SPECIFICATION

01 Master System Architecture

Master System Topology & Decoupled Trust Model • GovPulse Technical Architecture Suite • Version 1.0 (Production)

Master System Architecture Document

Project: GovPulse Autonomous Digital Government Certificate Platform

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1. Executive Vision & Core Architecture

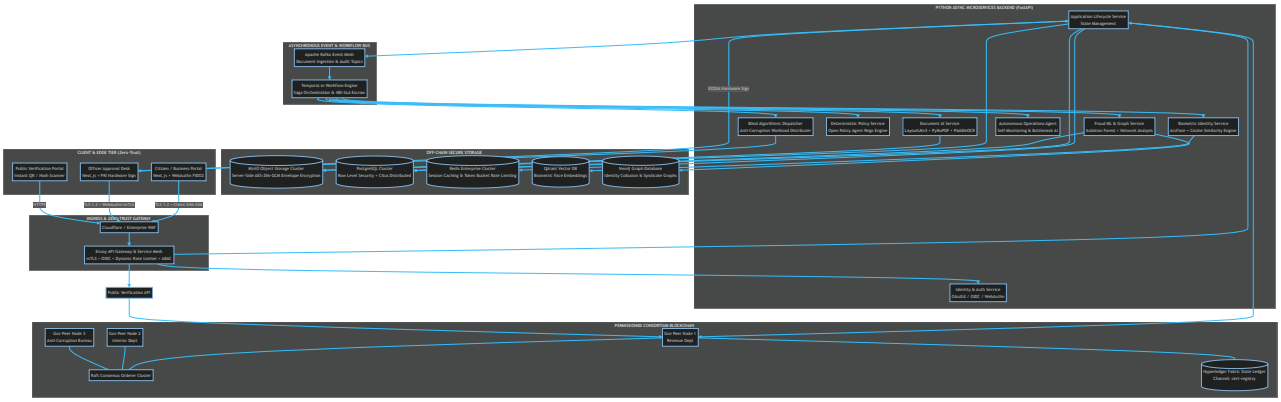
GovPulse is an enterprise-grade, zero-trust digital government operations platform designed to eliminate bureaucratic friction, human error, discretionary corruption, and multi-week processing delays in issuing official government certificates (e.g., Income, Domicile, Caste, Welfare, Disability, Trade Licenses).

The system replaces opaque, multi-layer manual clerk chains with an **autonomous, self-monitoring AI workflow engine** anchored by a **single accountable officer checkpoint** and a **permissioned blockchain proof registry**.

CORE ARCHITECTURAL TENETS
1. Decoupled Trust Model: Off-Chain AI & Encrypted Storage vs On-Chain Cryptographi
2. Strict Determinism Boundary: Probabilistic AI (Extraction) vs OPA Rego (Legal Co
3. Single-Officer Accountable Checkpoint with Zero Discretionary Opacity
4. Absolute Privacy (Zero PII On-Chain) with W3C Verifiable Credential Portability
5. Blind Algorithmic Dispatch & 48-Hour Strict SLA Escrow (Anti-Corruption Engine)

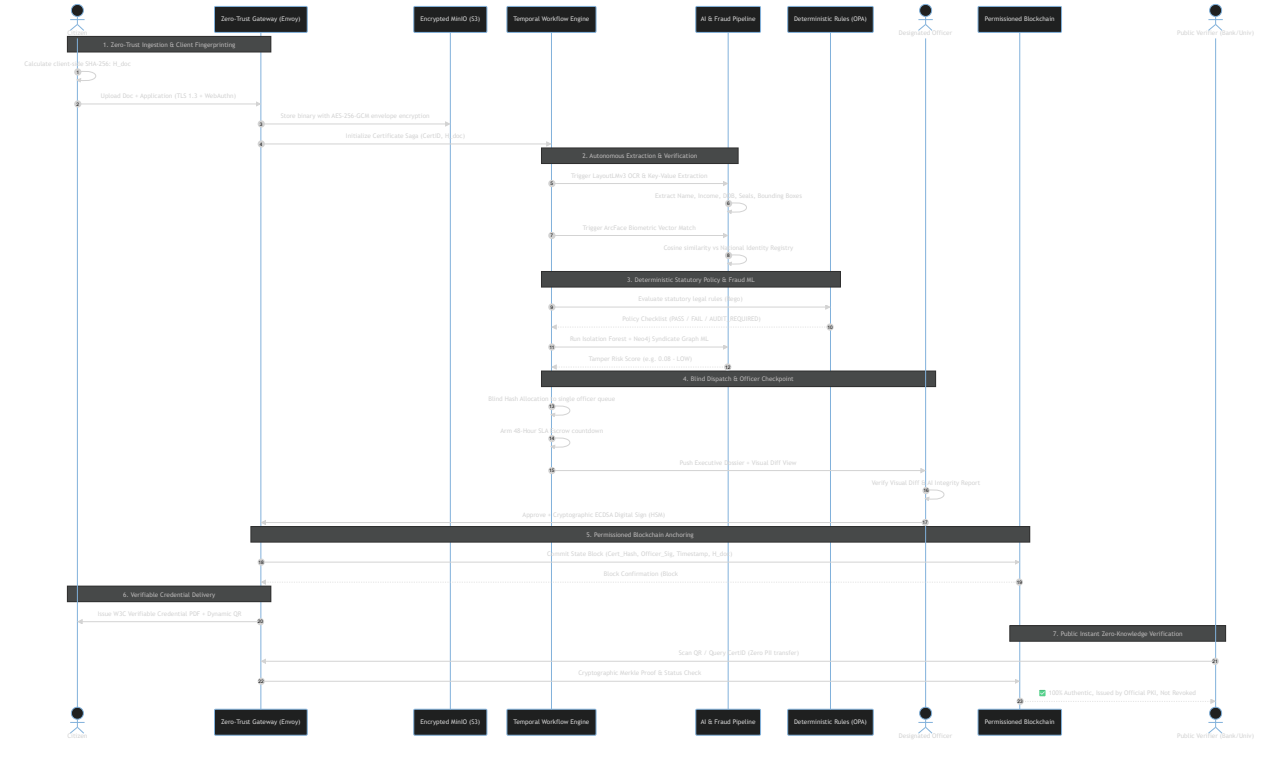
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2. High-Level System Architecture Diagram



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3. End-to-End Component Lifecycle



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4. Key Architectural Separation: Off-Chain vs On-Chain

To guarantee strict compliance with data privacy laws (e.g., GDPR, National Data Protection Acts), citizen personally identifiable information (PII) is **never** committed to the blockchain.

OFF-CHAIN LAYER (Confidential)	ON-CHAIN LAYER (Public Tr
• Full Scanned PDFs and Images • Citizen Names, Tax IDs, Addresses, Photos • Raw Financial & Income Figures • AI Extracted Layout Bounding Boxes • Temporary Processing States & Sagas • Neo4j Fraud Graph Relations Storage: MinIO AES-256-GCM, PostgreSQL, Qdrant Vector DB	• Document SHA-256 Fingerprint ($ • Unique Certificate Hash ($H_{ce • Officer PKI Public Key Fingerpr • Officer ECDSA Cryptographic Sig • Transaction Timestamp & Block M • Credential Lifecycle Status (VA Storage: Hyperledger Fabric Priva (State Database: CouchDB

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5. Architectural Quality Attributes & Non-Functional Requirements

METRIC	SPECIFICATION TARGET	ARCHITECTURAL ENABLER
System Throughput	10,000 Concurrent Submissions / min	Async Python FastAPI + Kafka Event Mesh + Redis caching
End-to-End Latency	< 45 seconds (Intake to Officer Queue)	Parallelized PyTorch GPU inference workers + OPA sub-millisecond evaluation
Public Verification Speed	< 800 milliseconds	Zero-Knowledge index lookup on Blockchain State Cache
High Availability	99.99% Uptime (Annual downtime < 52 mins)	Multi-AZ Kubernetes cluster + Raft consensus across 3+ validator peers

METRIC	SPECIFICATION TARGET	ARCHITECTURAL ENABLER
Data Integrity & Non-Repudiation	100% Mathematically Proven	Client-side SHA-256 + Officer HSM ECDSA signatures + Blockchain Merkle Proofs
Discretionary Bribery Index	0.00 (Mathematically impossible)	Blind hash dispatch + 48h SLA Escrow + Codified rejection requirements